APICC

Real-Time Data Pipeline for Smart Irrigation Systems

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Sap flow sensor

INTRODUCTION



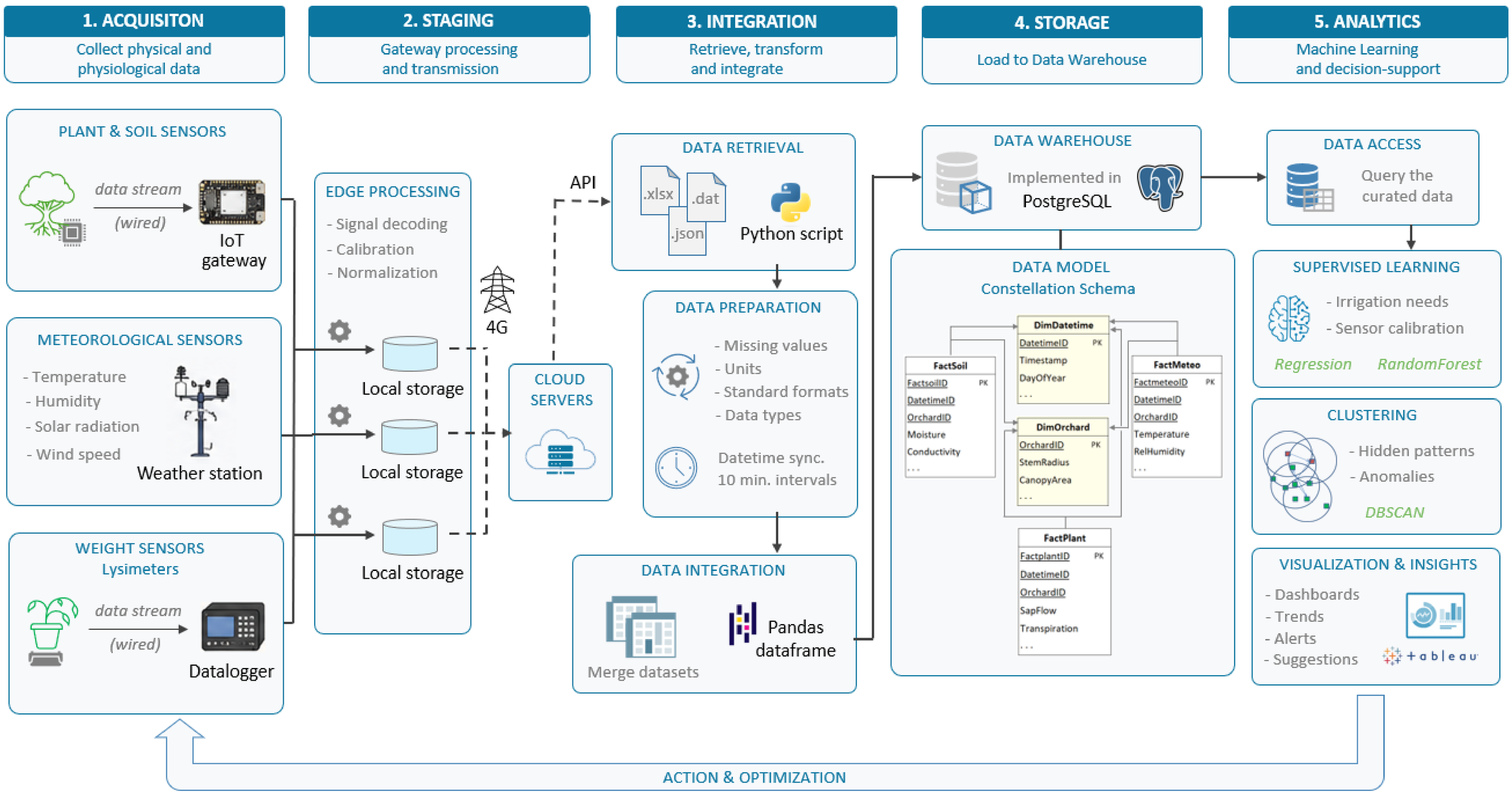
- An efficient irrigation system requires an accurate management.
- Optimization of irrigation systems involves continuous real-time monitoring of crop conditions.
- Deployed sensors generate vast amount of data that must be properly handled to produce actionable insights.
- The proposed data pipeline was applied for an olive orchard (*Olea europaea*) irrigation system.

DATASET

- The sensors measure soil, plant and environmental parameters.
- These heterogeneous variables exhibit different temporal granularities and do not share the same measurement time span.

	Plant Data	Soil Data	Meteorological Data
Sensors and Devices	• Sap flow sensors (SFS) • Lysimeters • Temp&Humi&Barometer sensor	• Capacitive soil moisture sensors (SMS) • Peltier cells	• Weather station
Parameters	• Sap flow velocity ($cm.h^{-1}$) • Plant transpirarion (mL) • Temperature at canopy (ºC) • Humidity at canopy (%)	• Moisture ($m3.m^{-3}$) • Heat flow ($W.m^{-2}$) • Electrical conductivity ($mS.cm^{-1}$)	• Air Temperature (ºC) • Relative Humidity (%) • Solar Radiation ($W.m^{-2}$) • Wind Speed ($km.h^{-1}$) • Vapor Pressure Deficit (kPa)
Granularity	1 minute	30 minutes	10 minutes

DATA PIPELINE METHODOLOGY



1. Acquisition: multiple sensors continuously collect physical and physiological data and transmit them to their respective gateways. Plant and soil sensors communicate via wired connections to a LIlYGO board, whereas the lysimeters, which measure variations in the orchard's weight to estimate transpiration, communicate with a CR3000 datalogger. Meteorological data are collected by sensors embedded in the weather station located near the olive grove (≈1 m above canopy).

2. Staging: raw signals received by the gateways undergo lightweight processing to convert them into interpretable measurements. The processed data are then transmitted via a 4G communication network to their respective cloud servers.

3. Integration: a local Python program automatically retrieves the data stored on the cloud servers through their APIs. The acquired data undergo preprocessing, including data cleaning, removal of missing values, transformation, and synchronization of timestamps. Finally, the heterogeneous datasets are integrated into a common tabular structure using a Pandas DataFrame.

4. Storage and Modelling: The integrated data are loaded into a PostgreSQL data warehouse organized as a constellation schema, with fact tables for soil, plant, and meteorological parameters sharing the DimTime (10-minute timestamps) and DimOrchard (tree characteristics) dimensions.

5. Analytics and Visualization: The curated data are retrieved from the PostgreSQL data warehouse and analyzed using machine learning techniques. Ensemble models estimate irrigation requirements, regression models calibrate sensors, and clustering algorithms identify hidden patterns and anomalies. The results are presented through interactive Tableau dashboards to support informed irrigation management.

CONCLUSION & FUTURE WORK

- This methodology enables scalable, low-cost, automated real-time irrigation management through efficient integration and analysis of heterogeneous data.
- Future work will extend the proposed methodology to other orchard and woody crops to evaluate its generalizability under different cultivation conditions.

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Project webpage: